Pre-Processing of Remotely Sensed Data

Image Restoration

Noise Removal/ Missing Scan Lines/ Dropouts

Cosmetic Operations

- **Correction of images with Bad Pixels or Missing Scan Lines (partially or complete) or line drop out (mostly in across track scanners)**
  - **Black/ White pixels (Random)**
  - **Horizontal black or white lines represented by a sequence of zero or 255 pixel (8 bit) values**

Cosmetic Operations

- **Bad Pixels or Missing Scan Lines Occurs**
  - **Due to errors in the scanning or sampling equipment**
    - **when a detector either completely or partially fails to function, during a scan**
  - **In the transmission or recording of the image data**
  - **Reproduction of the media containing such data , CD or tapes**
- **Affects the visual interpretation as well as statistical calculation based on pixel values**

Random Bad Pixels (Shot Noise)

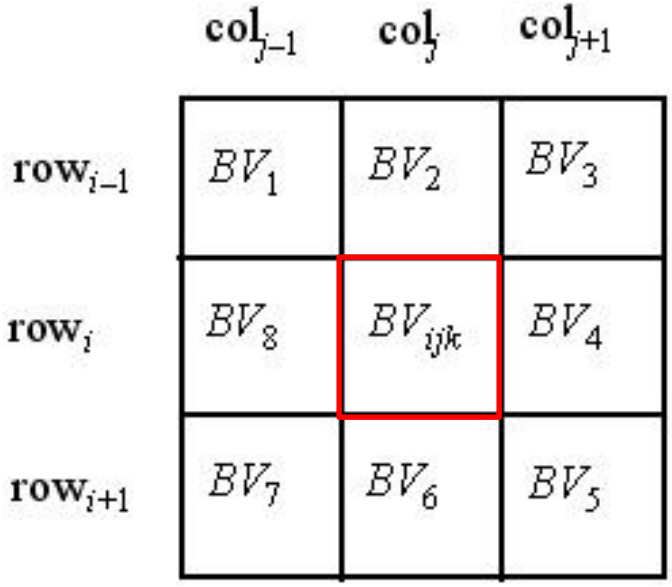
- **Sometimes an individual detector does not record spectral data for an individual pixel**
- **When this occurs randomly, it is called a *bad pixel***
- **When there are numerous random bad pixels found within the scene, it is called *shot noise* because it appears that the image was shot by a shotgun**
- **Normally these bad pixels contain values of 0 or 255 (in 8-bit data) in one or more bands**

Random Bad Pixels (Shot Noise)

- **Shot noise is identified and removed using the following methodology**
  - **First necessary to locate each bad pixel in the band *k* dataset**
  - **A simple thresholding algorithm makes a pass through the dataset and flags any pixel ($BV_{i,j,k}$) having a brightness value of 0 or 255 (assuming values of 0 or 255 represent shot noise and not real land cover such as water)**

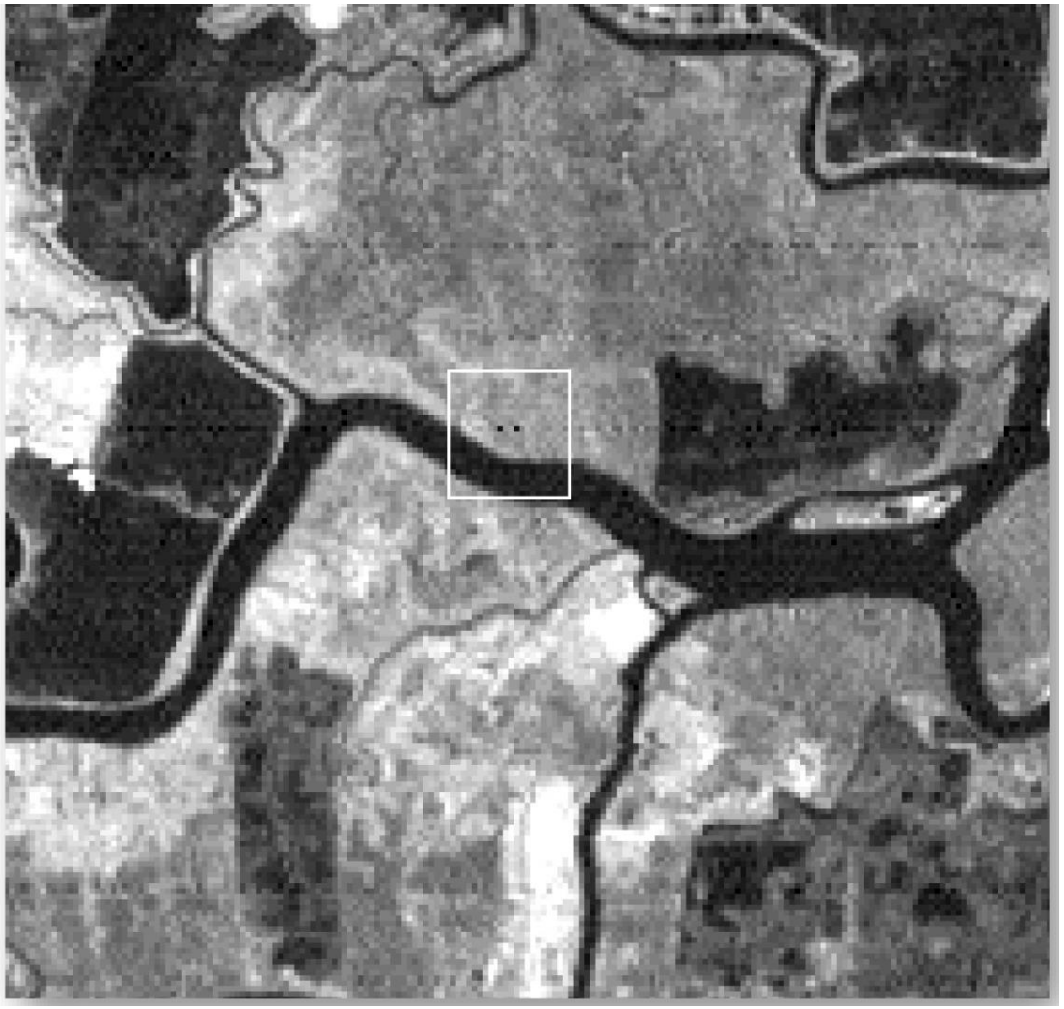
Random Bad Pixels (Shot Noise)

- **Once identified, it is then possible to evaluate the eight pixels surrounding the flagged pixel, as shown below:**

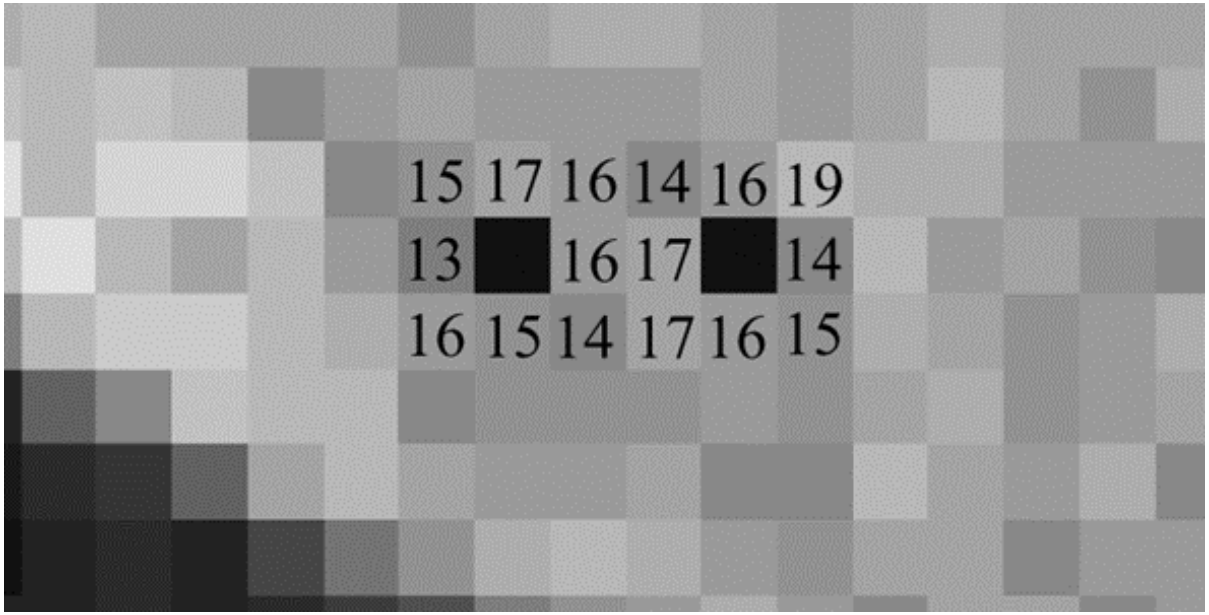


- **The mean of the eight (*n*) surrounding pixels is computed and the value substituted for $BV_{i,j,k}$ in the corrected image**

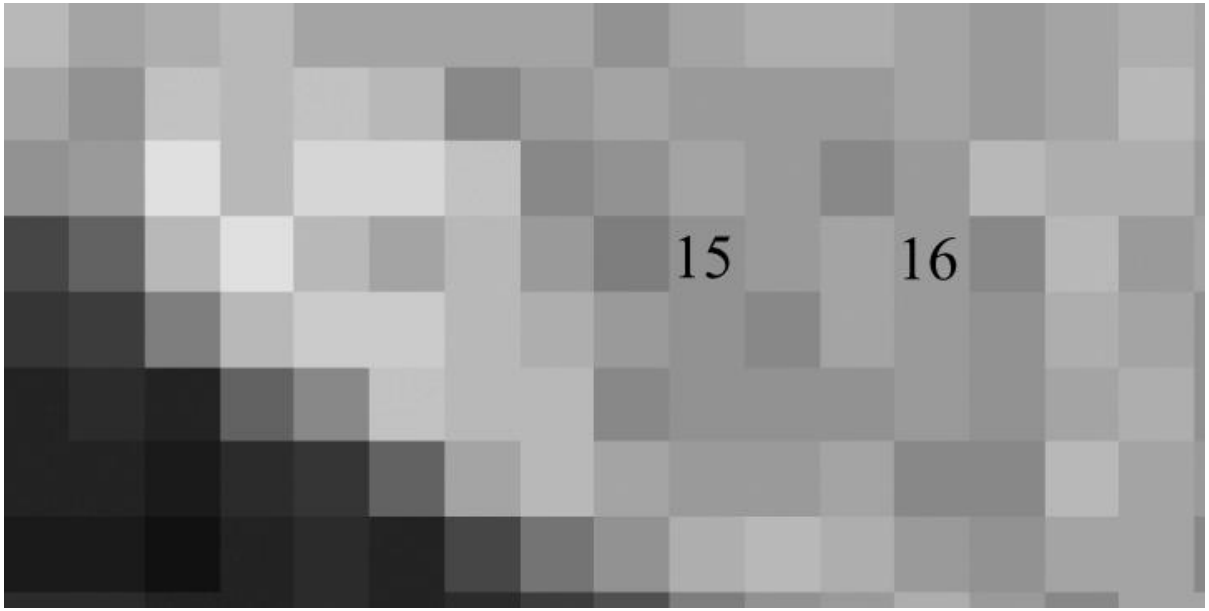
SHOT NOISE



SHOT NOISE



Before



After

Missing Scan Line, or Column Drop-outs

- **An entire line containing no spectral information may be produced if an individual detector in a scanning system fails to function properly**
- **If a detector in a linear array fails to function, this can result in an entire column of data with no spectral information**
  - **In Push broom system, each pixel in any one line is observed by an individual detector and a faulty detector in this system may show itself by a column of pixels with DN = 0**

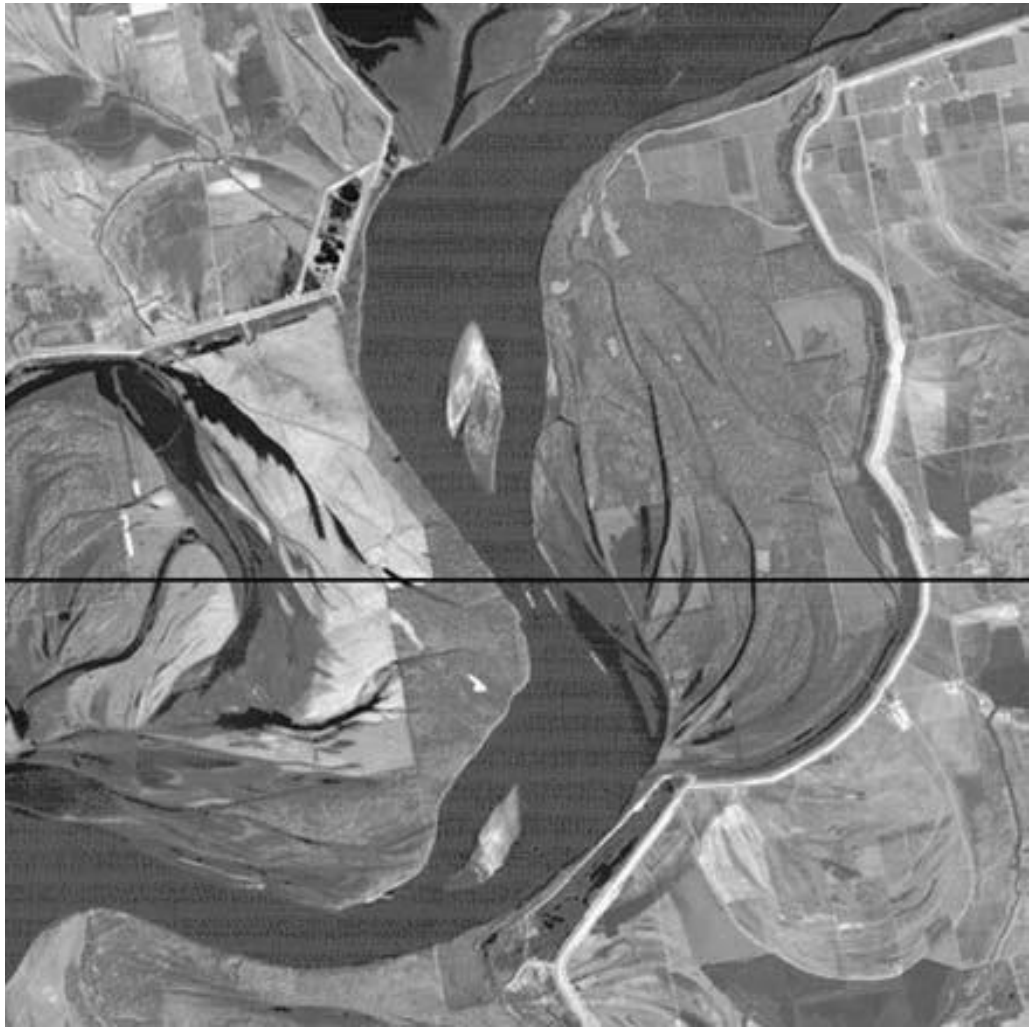
Line or Column Drop-outs

- **The *line drop-out* would appear as a completely black line in the band, *k*, of imagery**
- **This is serious problem because there is no way to restore data that were never acquired**
- **However, it is possible to improve the visual interpretability of the data by using estimated brightness values for each bad scan line**

Missing Scan Line

- **No means to know what data /information have been missed**
- **Just an effort to estimate what the missing values might be, by looking at image data values in the scan lines above and below**
- **Approach relies on " Spatial data autocorrelation "property**
- **Spatial auto correlation:**
  - **Point close in geographical space tend to have similar values**

Missing Scan Lines or line drop out





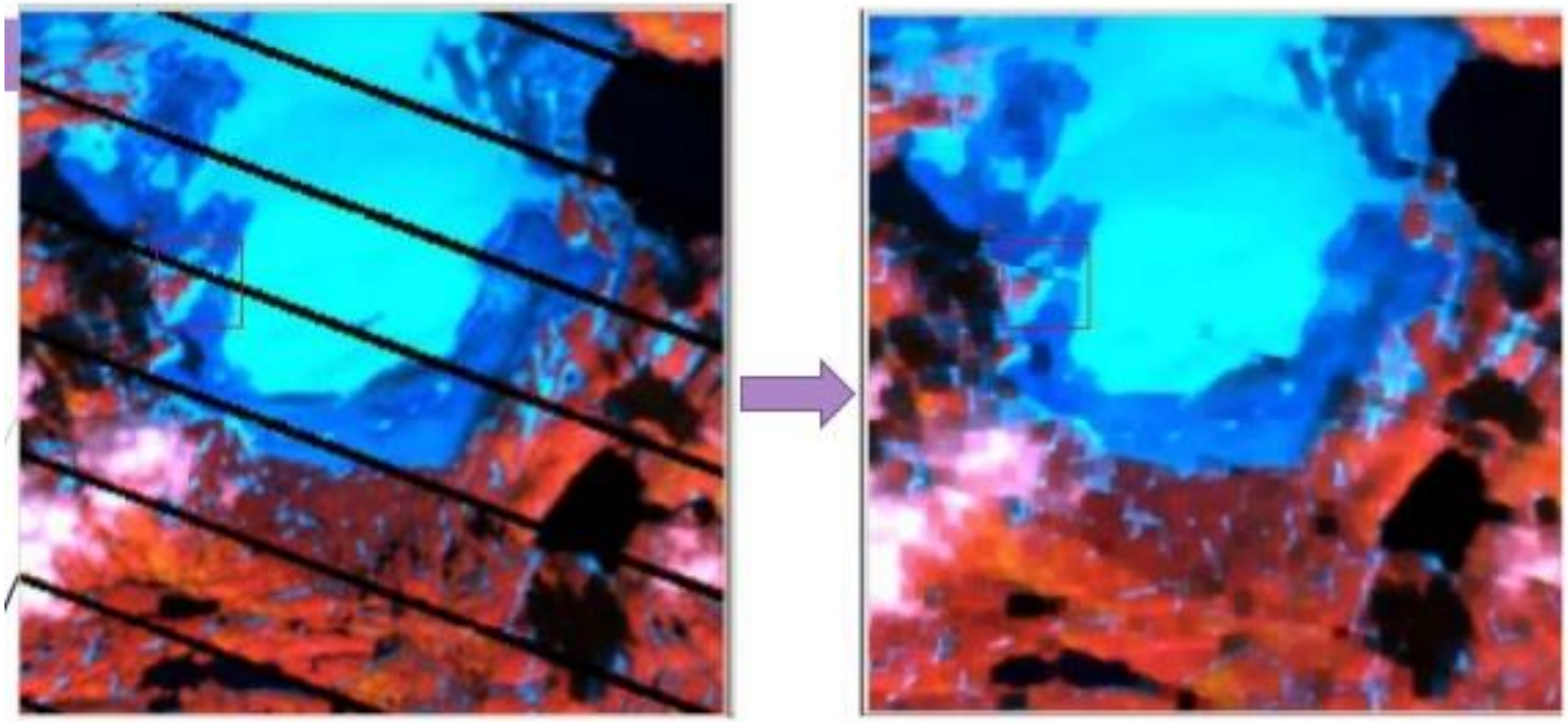
Missing Scan Lines or line drop out





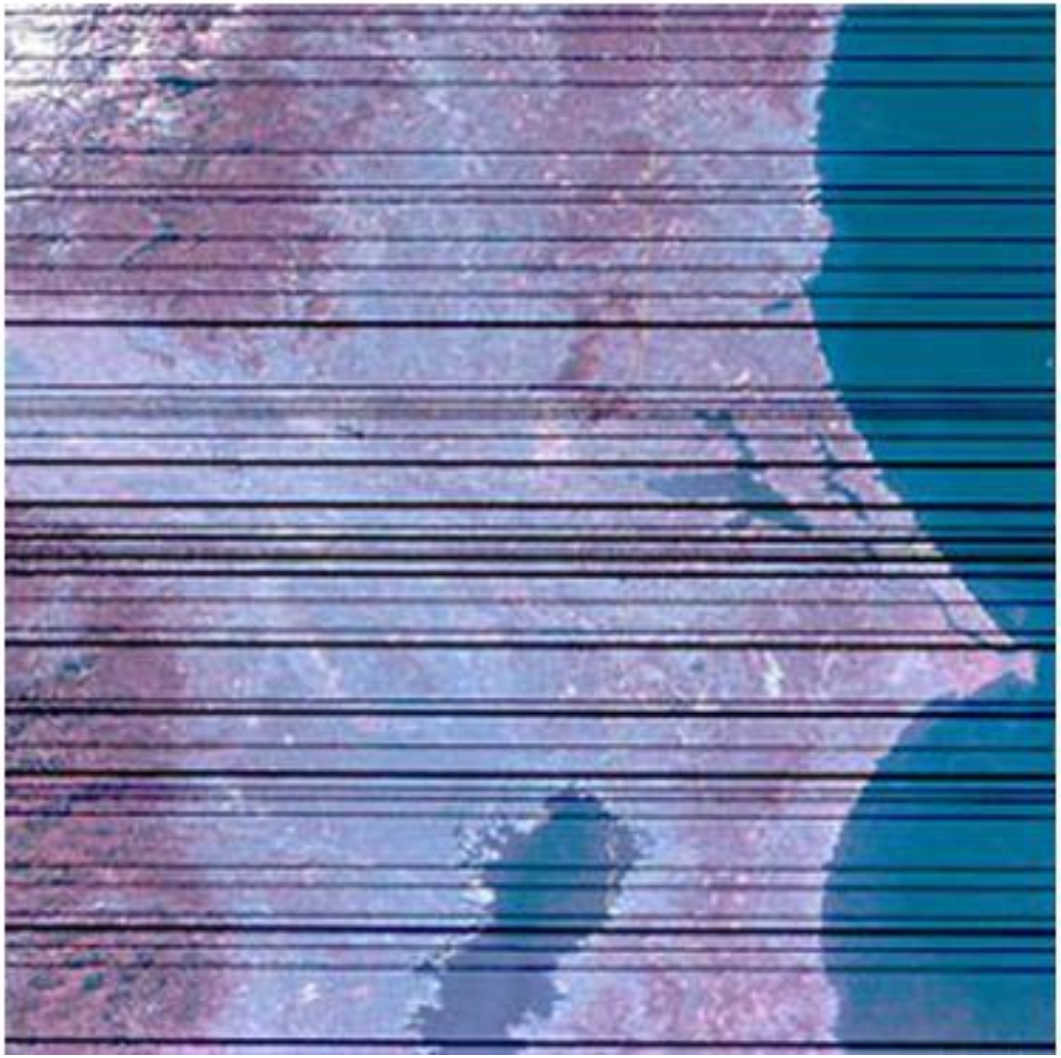
34	27	20	17	17	19	20	21	22	21	19	18	16	17	22
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
23	17	14	14	15	16	15	15	17	16	13	12	14	19	28
17	14	13	12	13	12	11	12	14	13	11	12	17	25	34
13	12	12	11	11	11	10	12	16	15	13	16	23	31	40
11	11	11	11	11	11	12	15	22	23	21	24	31	39	46
10	11	11	11	11	12	16	20	28	31	29	32	39	46	51
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
10	11	11	13	17	23	30	36	42	45	45	48	51	55	57
10	11	13	17	25	33	40	45	47	48	48	50	53	55	57
15	18	21	26	33	40	47	50	50	50	50	51	54	55	57
25	31	36	40	42	46	49	50	50	50	50	51	54	55	57
33	40	45	47	48	49	51	51	50	50	50	52	54	55	57
39	44	47	49	50	51	53	53	51	50	50	52	54	55	57
43	47	49	50	51	52	54	54	52	51	51	51	53	55	57
46	48	50	51	53	54	55	54	52	51	51	51	53	55	57
48	50	52	53	55	56	56	55	53	51	50	50	52	55	57
50	52	54	55	57	58	58	56	54	51	50	50	52	55	58
51	53	55	57	58	59	59	57	55	52	49	49	51	54	59
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
53	55	57	59	58	58	58	57	55	52	48	49	52	55	61

Missing Scan Lines or line drop out



With a pattern-Systematic, Landsat 7 Satellite

Missing Scan Lines or line drop out



With no pattern-Un-Systematic

Missing Scan Line

- **Estimation of missing values from a scan line from the adjacent values**
- **It is corrected by replacing the bad line with a line of estimated data file values, which are based on the line above or below it**

Missing Scan Line

- **Method-1(simplest)**
- **Estimating a missing scan line values and the replacement by the corresponding values from the immediately preceding scan line i.e.**
  - **$V_{i,j,}$ =Value of Pixel i on scan line j**
  - **Then the algorithm is simply**
    - **$V_{i,j,} = V_{i,j-1}$**
  - **Simple method, ensure the replacement value from nearest neighborhood of pixel (i, j) , similar to at least one of its neighbor**
  - **When missing line(j) is first line, then the replacement could be**
    - **$V_{i,j,} = V_{i,j+1}$**

Missing Scan Line

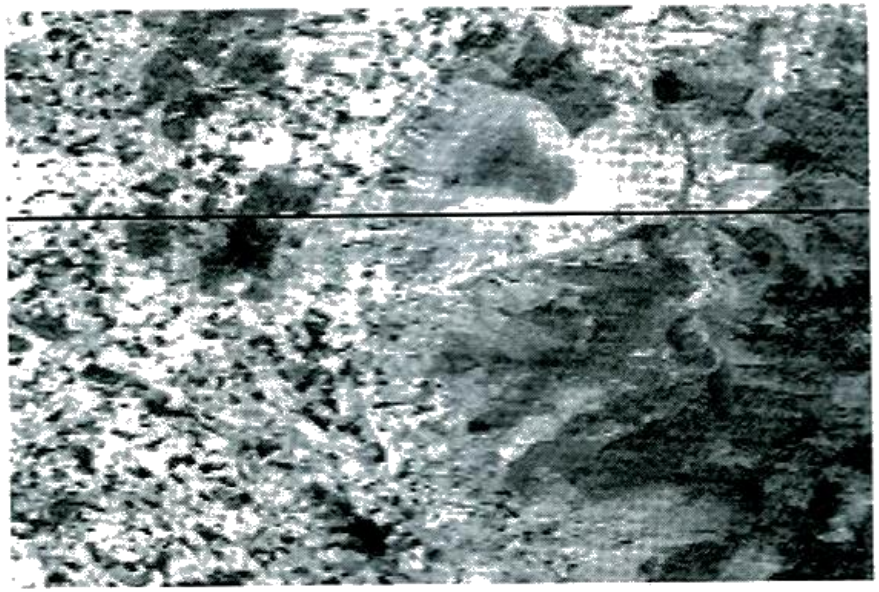
- **Method-2**
- Replacing the missing values by the average of the corresponding values of the pixels on the scan lines immediately above and below the defective line, i.e.
  - $V_{i,j,}$ =Value of Pixel i on scan line j
  - $V_{i,j,=}(V_{i,j-1} + V_{i,j+1})/2$
  - When missing line(j) is first line or last , then the replacement could be done by method 1.
  - When the nearby pixels values are not highly correlated, this method produces pixel value unlike neighbors, a case where the missing line may be the boundary of two different land covers (e.g. between land and water)

Missing Scan Line

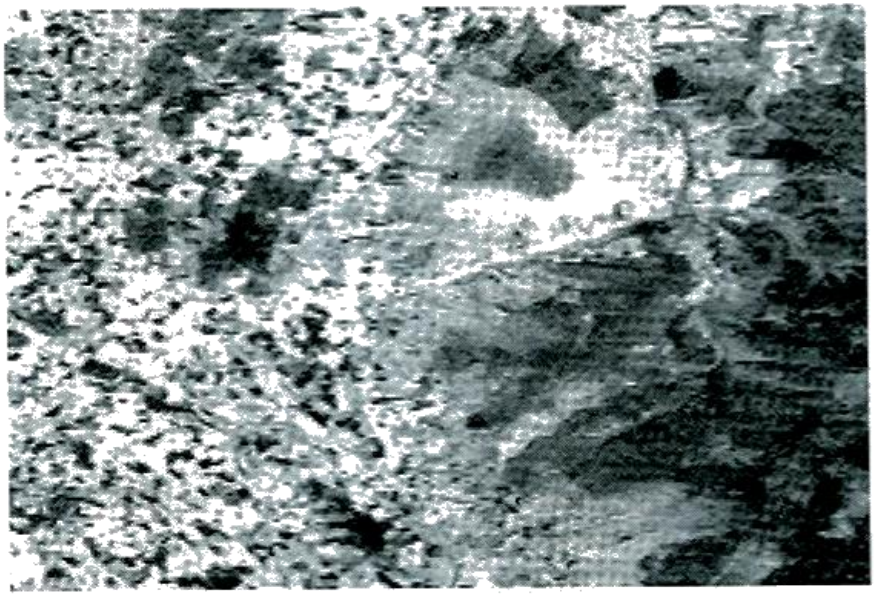
- **Method-3 (complex)**
- Relies that two or more bands of imagery are available
- Uses those two bands that are highly correlated
- Replacing the missing pixel values in band **K** by estimating the contribution from
  - The equivalent pixels from another highly correlated band **r**
  - Neighboring pixel in the same band (Method 1,2)
  - $V_{i,j,k} = M\{V_{i,j,r} - (V_{i,j+1,r} + V_{i,j-1,r})/2\} + (V_{i,j+1,k} + V_{i,j-1,k})/2$

- Where M is the ratio of the std deviations of pixel values on band K and r

Missing Scan Line



a



b

Locating

- Compute average of the pixel values along scan line i to n (n=no of scan lines in image)
- Expected that the average of line i+1 and i-1, differ from the average of line I (the missing scan line)
- Compute the difference (d) between the average of line i+1 and i-1,
- Check all scan line for those that have deviation more than (d)
- Check a few such lines pattern for conformation